Solve the following LP problems using your preferred software, cleary explaining and stating the variables to be optimized, the objective function and the possible constraints of the problem. Upload to CANVAS the workbook used to solve the problem.

Problem 1:

A logistics company operates a fleet of three types of delivery trucks: Small, Medium, and Large. Each truck type has different costs, capacities, and fuel efficiencies. The company aims to optimize the fleet composition to maximize delivery efficiency (total capacity) while minimizing operating costs. They have limits on the total number of trucks they can operate due to parking space limits to only 60 trucks and budget for operating costs of $10,000.

Small Truck:

Capacity: 600 kg

Operating cost per day: $85

Fuel efficiency: 15 km/l

Maximum trucks available: 30

Medium Truck:

Capacity: 1300 kg

Operating cost per day: $199

Fuel efficiency: 10 km/l

Maximum trucks available: 30

Large Truck:

Capacity: 2000 kg

Operating cost per day: $301

Fuel efficiency: 5 km/l

Maximum trucks available: 20

Problem 2:

Caterpillar is a leading grower and distributor of fresh citrus products with three large citrus groves scattered around central Florida in the cities of Mt. Dora, Eustis, and Clermont.

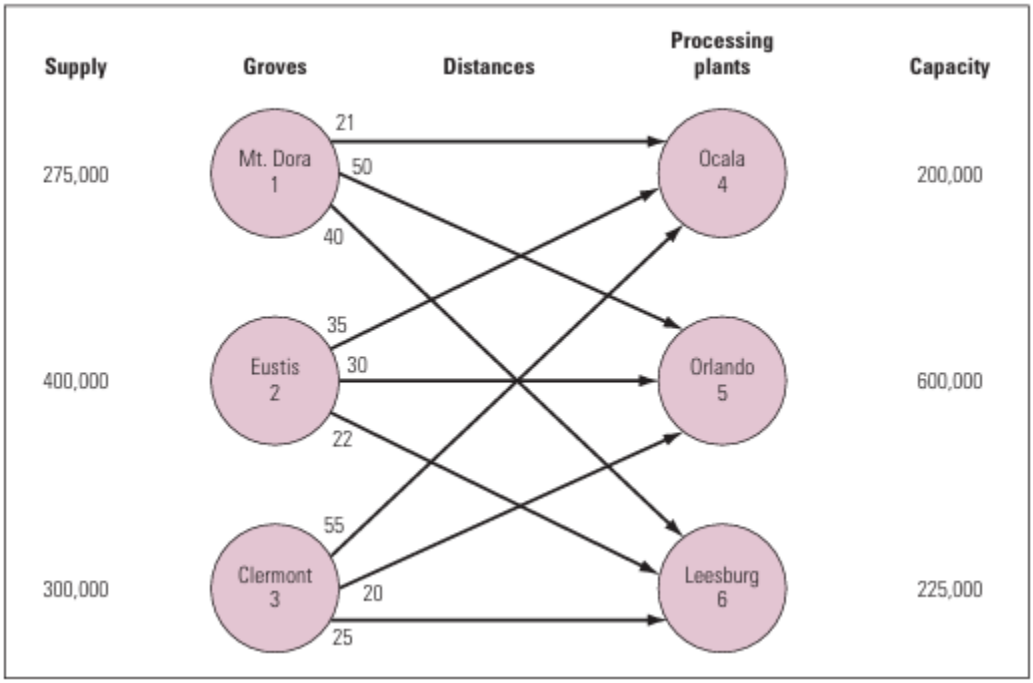
Caterpillar currently has 275,000 bushels of citrus at the grove in Mt. Dora, 400,000 bushels at the grove in Eustis, and 300,000 bushels at the grove in Clermont. Caterpillar has citrus processing plants in Ocala, Orlando, and Leesburg with processing capacities to handle 200,000, 600,000, and 225,000 bushels, respectively.

Caterpillar contracts with a local trucking company to transport its fruit from the groves to the processing plants. The trucking company charges a flat rate for every mile that each bushel of fruit must be transported. Each mile a bushel of fruit travels is known as a bushel-mile.

The following table summarizes the distances (in miles) between the groves and processing plants: Distances (in miles) between Groves and Plants Grove Ocala Orlando Leesburg Mt. Dora 21 50 40 Eustis 35 30 22 Clermont 55 20 25 Caterpillar wants to determine how many bushels to ship from each grove to each processing plant in order to process all the fruit while minimizing the total number of bushel-miles the fruit must be shipped.

	Distances (in miles) between Groves and Plants		
Grove	**Ocala**	**Orlando**	**Leesburg**
Mt. Dora	21	50	40
Eustis	35	30	22
Clermont	55	20	25

In this situation, the problem is to determine how many bushels of fruit should be shipped from each grove to each processing plant. The problem is summarized graphically



Exercise 1

Plot the feasible region in 3D from the problem below; considering the x axis as the units of product A, y axis as the units of product B and the z axis the total revenue.

You're the manager of a company that produces two types of products: Product A and Product B and you are planning the monthly earnings budget. Each of your products requires a certain amount of two resources: labor hours and raw materials. However your company only works 100hrs this month and the available raw material is 90kg.

For the product A: 8 labor hours and 2 kg of raw material are needed and the revenue is $40 USD per unit.

For the product B: 1 labor hour and 2 kg of raw material are needed and the revenue is $30 USD per unit.

Your goal is to determine the optimal production mix of Product A and Product B to maximize profit while staying within the resource budget.